

SYSTEM / DP-SGD

Secret: candidate absent or eligible
for sampling at rate q each round

Clip private batch gradients at θ_t
Release noisy average y_t

Update $\theta_{t+1} = \theta_t - \eta y_t$; repeat

Hidden: batch, inclusion bits, noise draws

visible
 (θ_t, y_t)

ADVERSARY / GAUSSPROOF

Known: candidate z , public D_{pub} ,
model and DP-SGD settings

At θ_t , recompute h_t and b_t
from candidate and public data

Compare $y_t - b_t$ with shift $(h_t - b_t)/B$

Accumulate q -aware evidence $M_T(z)$
Test membership or rank a gallery